

5 One cause of blindness is a mutation in the DNA coding for the enzyme RPE65.

The enzyme RPE65 converts trans-retinal to cis-retinal.

(a) Explain why mutations in the RPE65 gene can result in blindness.

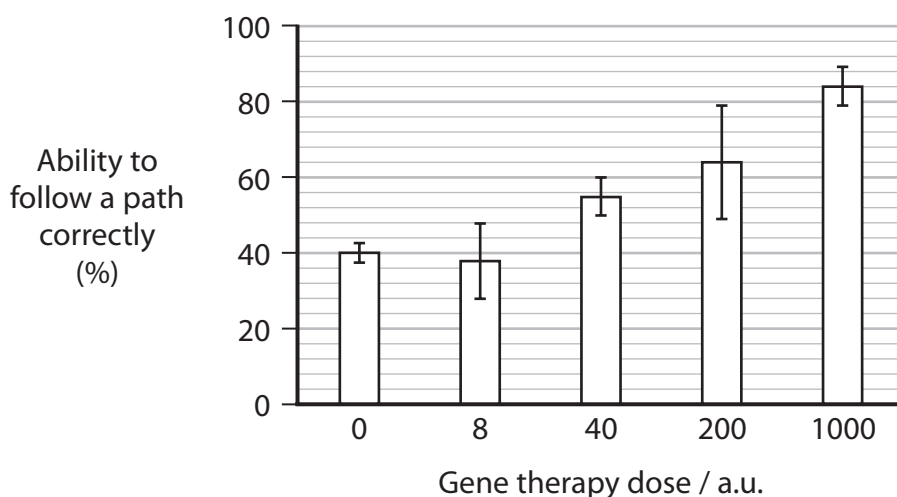
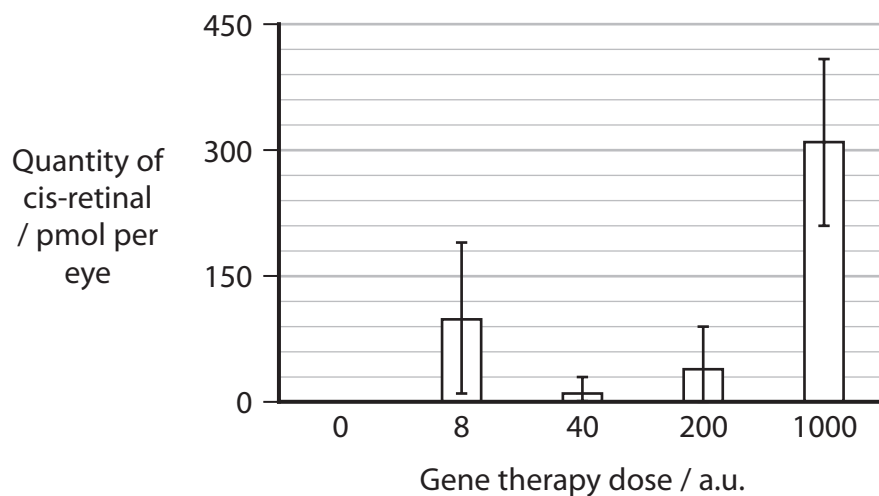
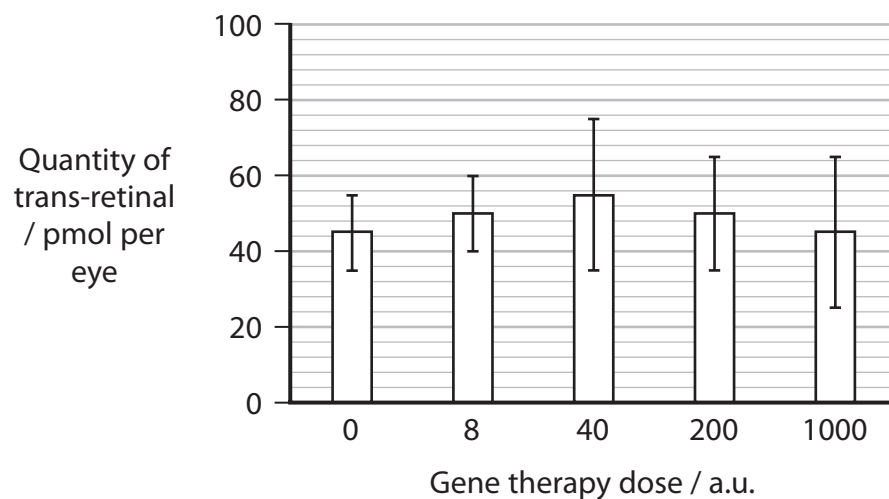
(3)

- (b) Scientists are investigating the use of gene therapy to treat blindness caused by mutations in the RPE65 gene.

In one investigation, different gene therapy doses containing functioning RPE65 genes were compared.

Two months after treatment, the trans-retinal and cis-retinal in each eye and the ability to follow a path were measured.

The graphs show the results of this investigation.



Use the information in the graphs to support your answer.

[illegible]

(c) Mutations in other genes can result in blindness.

Describe how these genes could be identified.

(3)

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(Total for Question 5 = 10 marks)